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PIYON RAJ

Unity Developer (AR&VR)

Education

Bsc Computer science
Alagappa University
2019-2022

HSC Bio Maths
Govt higher Secondary School
2017-2019

Language

English
Tamil
Kannada
Malayalam

Tools & Software

- Html
- CSS
- C#
- Excell
- Unity
- Photon
- React js
- Photoshop
- Git&GitHub
- Blender
- Unreal
- Audacity
- Articulate 360 / Storyline
- ChatGPT Codex
- Antigravity
- WebXR
- UVC (DevOps)
- Unity Addressables
- HoloLens (MR / AR)
- Unity MCP

Profile

Hi, I'm piyonraj — a passionate Unity Developer with 2.9+ years of hands-on experience creating immersive, interactive experiences.

I specialize in game design, Unity, C# programming, and XR development, bringing ideas to life with precision and creativity. From dynamic gameplay systems to cutting-edge AR/VR features, I focus on building polished, performance-driven projects that captivate users and meet real-world goals.

Work Experience

2023/Aug – 2025/Jan

GI Tech Private Limited ,Chennai
Unity Developer

- Collaborated with cross-functional development teams to build scalable Unity applications.
- Designed 2D/3D assets and UI/UX interfaces powered by clean C# scripting.
- Implemented TCP connections and managed middleware data handling for robust system communication.

2025/Jan – 2025/Apr

V4C Solutions , Kerala
HAL , Bangalore (Contract 3 months)
VR Developer

- Working with Modular and whole team.
- 2D & 3D , UI&UX Designing and Simulations With c# Script.
- XR Development Plugins with Oculus
- Identify Bugs and resolve in VR Development.
- Realistics, Physics, Rigidbody Interaction.

2025/Aug – Present

Karanji Info Tech , Mangalore Karanataka
VR Programmer

- **Advanced XR & SDK Integration:** Engineered production-ready VR applications by implementing Meta XR SDKs, All-in-One SDKs, and custom XR packages, executing seamless hand-tracking, gesture recognition, and cross-platform hardware integration.
- **Complex Mechanics & Physics:** Architected hyper-realistic virtual reality simulations featuring advanced Rigidbody interactions, physics-based mechanics, and responsive hand-tracking for immersive user interaction.
- **Modular Development & Scripting:** Built scalable, modular VR architectures and intuitive UI/UX workflows utilizing robust C# scripting, ensuring clean codebases and rapid feature deployment.
- **Full-Cycle Optimization & Debugging:** Diagnosed and resolved critical performance bottlenecks, optimizing frame rates, rendering pipelines, and stability across complex multi-user VR environments.

Skills

- Unity & C# Engineering: Advanced C# scripting, Object-Oriented Programming (OOP), modular architecture, and robust gameplay mechanics.
- XR & Spatial Computing (VR/AR): Oculus/Meta XR SDKs, OpenXR, VRIF (Virtual Reality Interaction Framework), hand-tracking, gesture recognition, Vuforia, and ARCore.
- Physics, Animation & Interactions: Complex Rigidbody physics, realistic environmental interactions, and animation tweening (DOTween, ITween).
- Multiplayer & Networking: Real-time multiplayer implementation and network synchronization using Photon (PUN 2) and XR networking pipelines.
- Performance Optimization & Debugging: Frame-rate profiling, memory optimization, draw-call reduction, and rigorous cross-platform debugging.
- Cross-Platform & Pipeline Workflow: Deployment across PC, Mobile, and VR Headsets; Git version control; rapid prototyping; and direct 2D/3D asset workflow integration.

Project Summary

Casino Games - **GI Tech Private Limited**

Is all of Online 2d&3d Betting Games , that games included functionality's are - login, physics, Colliders, animation controllers , UI&UX designing , Plugins, Socket connection and dynamic result functionality with server data.

- **Gulf Ticket**
  - Super 6
  - Fortune 4
  - Lucky 1
  - Dream 5
- **Taj Draw**
  - Joker
  - Ace Joker
  - Dream 5 & Lucky 4 & Pick 3
  - Taj Million

Virtual Reality - **V4C Solutions**

- **Tejas - LCA 1**

  Developed 7 defense training modules for the Indian Air Force (HAL) featuring realistic physics, Oculus controls, and guided step-by-step procedures for aircraft parts removal, tool handling, and panel maintenance.

Virtual Reality - **Karanji Info Tech**

1. Medverse (Healthcare, WebXR & SCORM)

- **What it is:** A 5-module immersive healthcare training suite built in Unity using VRIF and BNG frameworks.
- **Key Highlights:** Built a centralized application launcher, integrated hand-tracking for select modules, and deployed WebXR builds compatible with SCORM compliance for seamless LMS/browser delivery.

2. L&T Industrial Simulation (Three-Phase Induction Motor)

- **What it is:** An interactive virtual reality electrical training simulation developed specifically for Larsen & Toubro (L&T).
- **Key Highlights:** Focused on equipment assembly, maintenance workflows, and electrical circuit testing (specifically testing current supply on a Three-Phase Induction Motor) using VRIF controller mechanics.

3. Spiral Cage Sewing Machine Simulation

- **What it is:** A specialized virtual reality training simulator for industrial sewing machine workflows.
- **Key Highlights:** Replicated mechanical assembly, operational steps, and maintenance procedures using realistic VRIF controller mechanics. Integrated Unity Addressables and Asset Bundles for memory optimization and dynamic asset.

4. CNC Machine Virtual Training Simulator

- **What it is:** A comprehensive manufacturing simulator for precision CNC machine operation.
- **Key Highlights:** Replicated real-world production steps including raw material setup, value configurations, program parameter setting, precision grinding workflows, and component removal.

5. Dental VR Training Simulation (Current Project)

- **What it is:** An advanced medical training simulation for dental procedures.
- **Key Highlights:** Supporting dual interaction modes (Controller-based and high-precision Hand Tracking via Meta XR / VRIF) to train medical professionals in procedural accuracy.

6. AI-Assisted Quiz Web Platform

- **What it is:** A fully functional interactive quiz web application.
- **Key Highlights:** Designed and developed using modern web workflows and AI-assisted coding tools (AIDE / AI Codex), showcasing your ability to build web solutions outside of Unity.

INDEPENTEND GAMES & APPS

Forklift Driving (unity)

Platform: Windows
Engine: Unity 6000.0.47f1
Role: Solo Developer
Studio: Omega Studios
Duration: [Apl 2025 – Apl 2025]
Project Overview:
A realistic 3D forklift driving simulation game designed to mimic the mechanics and physics of industrial forklift operation. The game challenges players to navigate an industrial environment and operate the forklift hydraulics to move and place objects precisely.
Key Features:

- Vehicle Physics: Implemented realistic physics using Unity's Rigidbody and Collider components to simulate weight and motion.
- **Control Scheme:**
  - W / S: Move forward and backward
  - A / D: Steer left and right
  - I / K: Control hydraulic lift (up and down)
- **Environment & Assets:**
  - Forklift model from Unity Asset Store (Low Poly Forklift)
  - Industrial environment from Asset Store (RPG_FPS_game_assets_industrial)
- User Guidance: On-screen instructions to guide new users through controls and objective-based tasks.
- Target Platform: Built for Windows PC.

Responsibilities:

- Integrated 3D models and configured environment setup.
- Created responsive forklift movement and lifting system using Unity physics.
- Designed user-friendly control layout and UI instruction prompts.
- Tested for smooth and realistic player interaction.

Primary Kids(unity)

Platform: Android
Engine: Unity 6000.0.47f1
Role: Solo Developer
Studio: Omega Studios
UI Tools: Unity 2D Canvas, UI assets designed in Photoshop CC 2023
Status: [it will be play store]
Project Overview:
An interactive and engaging educational app for kids to learn basic concepts like alphabets, numbers, shapes, and object names. Designed to combine visual learning with audio feedback and fun drawing activities, this app encourages self-learning for preschoolers and toddlers.
Key Features:

- **Learn with Audio:**
  - Learn A–Z, 1–10, shapes, animals, birds, objects, and common words like apple, ball, cat, horse, etc.
  - Click on the speaker button to hear the correct pronunciation of each item.
- **Dictation Page:**
  - A "Generate" button randomly selects a letter or number (based on the last visited learning section).
  - The child can listen to the generated item using the speaker icon, supporting audio-based recognition and memory recall.
- **Drawing Page:**
  - Free drawing space where kids can practice writing letters, numbers, or draw shapes.
  - Finger/mouse support for intuitive sketching.
  - Erase button to clear the screen and start over.

Design & Development:

- Fully built using Unity's 2D Canvas UI system.
- Custom UI assets created in Photoshop CC 2023 for a bright, child-friendly interface.
- Touch-friendly layout designed for mobile interaction on Android devices.
- Developed and published under Omega Studios brand on Play Store.

Responsibilities:

- Designed and developed the full learning flow, audio logic, and user interaction.
- Integrated touch-based drawing features and audio playback system.
- Created UI assets and ensured a consistent, colorful theme appealing to young children.
- Conducted testing for usability, performance, and accessibility on Android devices.

INDEPENTEND GAMES & APPS

Escape Dot(unity)

Platform: Android
Engine: Unity 2022.3.61f1
Role: Solo Developer
Studio: Omega Studios
Status: [It will be play store]

Project Overview:
Escape Dot is a minimalistic touch-based puzzle game where the player navigates a small dot through a blocked maze. The objective is to guide the dot from the bottom-left to the top-right escape point, overcoming physical obstacles and unlocking gates using interactive triggers.

Key Features:

- **Touch Controls:**
  - Simple swipe mechanics allow the player to move the dot up, down, left, and right.
  - Smooth mobile interaction built specifically for touch input.
- **Interactive Puzzle Mechanics:**
  - A valve key must be triggered to unlock a blocking obstacle.
  - Players must think and plan movements to reach the valve before accessing the escape.
- **Level-Based Progression:**
  - After escaping the current maze, players advance to the next level.
  - Increasing difficulty with each new stage to improve player engagement.
- **Design Philosophy:**
  - Clean, minimal design for a casual puzzle-solving experience.
  - Focus on simple interaction, logical reasoning, and rewarding progress.
  -

Responsibilities:

- Designed and implemented all game mechanics including touch movement, collision logic, key-trigger system, and level transitions.
- Built intuitive mobile controls for Android using Unity Input System.
- Optimized performance for mobile devices and managed level scaling.
- Created and tested level progression and puzzle balancing.

Bird Hunting(unity)

Platform: Windows (Planned Multiplayer)
Engine: Unity 6000.0.47f1
Role: Solo Developer
Assets Used: 2D Bird Sprites from Unity Asset Store (2DBirds_Pack3)
Status: complete for single player, In Development for Multiplayer.
Project Overview:
Bird Hunting is a 2D side-view shooting game where the player stands on a beach with a gun, aiming to shoot flying birds for points. The game focuses on timing, aim, and quick reactions, with plans for a future multiplayer mode for competitive fun.
Key Features:

- **Player Mechanics:**
  - Player is positioned on the ground with a fixed upward-pointing gun.
  - Movement is controlled using A / D or Left / Right arrow keys.
  - V key is used to shoot bullets directly upward.
- **Bird Spawning System:**
  - Birds fly randomly across the screen, spawning from 3 different random points.
  - Includes 3 types of birds, selected and spawned at random.
- **Scoring System:**
  - When a bullet hits a bird, the player earns 1 point.
  - Score tracking integrated with feedback for successful shots.
- **Environment Design:**
  - Set in a beach-themed background to create a calm yet exciting hunting atmosphere.
  - Clean 2D visuals optimized for desktop play.
- **Future Roadmap:**
  - Multiplayer mode planned using Photon or Nakama for real-time competition.

Responsibilities:

- Implemented 2D character movement, shooting mechanics, and collision detection.
- Integrated asset store prefabs and designed beach environment layout.
- Created randomized spawn logic and scoring system.
- Planning architecture for future online multiplayer extension.

INDEPENTEND GAMES & APPS

GPS Camera (unity)

Platform: Android
Engine: Unity 2022.3.61f1
Role: Solo Developer
Status: In Development (Testing & Updating in Progress)
Packages Used: Native Camera, Native Gallery (Unity Asset Store)
API Used: OpenCage Geocoding API
Project Overview:
GPS Camera is a custom mobile camera application that allows users to capture photos with embedded location information, including address, latitude/longitude, and timestamp. Designed for field use, documentation, or travel logs, the app combines camera functionality with real-time location data.
Key Features:

- **Permission Handling:**
  - Requests Camera, Location, and Storage permissions on app launch.
- **Live Camera Preview:**
  - Uses Native Camera package to display a live camera feed on screen.
- **Real-time Watermark Overlay:**
  - On the bottom-left, overlays the following info:
    - Full Address (via OpenCage API)
    - Latitude / Longitude
    - Date & Time with seconds
- **Capture and Save Image:**
  - On tapping the shutter button, the app captures the screen with watermark.
  - Image is saved to device storage at:
  - DCIM/GPS_CAMERA/CAPTURED_IMG1.jpg
- **File Organization:**
  - Custom folder created for saving captured images in organized format.
- **Use Cases:**
  - Useful for field workers, travelers, documentation, or location-based photo capture needs.

Responsibilities:

- Integrated camera preview and photo capture using Native Camera.
- Retrieved geolocation and full address using OpenCage API.
- Displayed real-time location and time data overlay using Unity UI.
- Managed Android file saving using Native Gallery and file path customization.
- Currently performing testing for address accuracy, permission stability, and save performance.

4 Wheel Fight

Engine: Unity 6 | Platform: PC | Type: 3D Multiplayer Racing
A 4-player online racing game using Photon PUN2. Players drive with WASD/Arrow keys and use Space to brake. Only 4 players per room; excess players join new rooms automatically.

Dragon Mario

Engine: Unity 6 | Platform: PC | Type: 2D Side-Scroller
A 2D runner with A/D or Arrow keys for movement and Space to jump. Features custom run and jump animations with sprite swapping.

Class Ender

Engine: Unity 2022.3.61f1 | Platform: PC | Type: 3D Endless Runner
A unique running game where the player's size changes based on points. Collect blocks to grow, avoid losing points to prevent shrinking. Win or lose based on final score.

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